

Constrained Delaunay tetrahedrisation refinement

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July 22, 2025

Outline

Introduction

Who am I?

Constrained Delaunay tetrahedrisation

State of the art

Quick presentation

- ▶ Max Royer
- ▶ ENS of Lyon
- ▶ End of first year of master
- ▶ Internship for 3 months



Figure: A ragu I made in Genova

Delaunay triangulation



Figure: A triangulation and a Delaunay triangulation in 2D

Delaunay triangulation

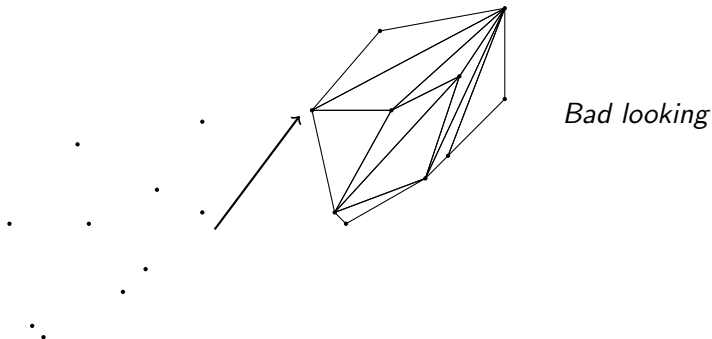


Figure: A triangulation and a Delaunay triangulation in 2D

Delaunay triangulation

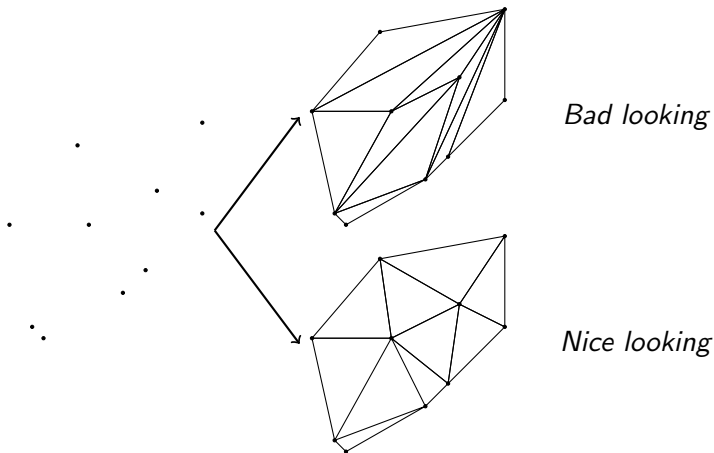


Figure: A triangulation and a Delaunay triangulation in 2D

Delaunay condition

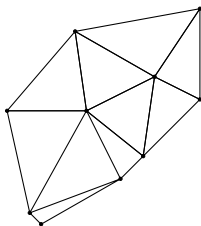


Figure: Example for Delaunay condition

Delaunay condition

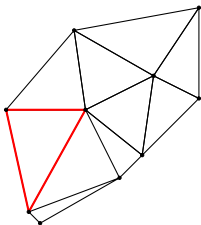


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Delaunay condition

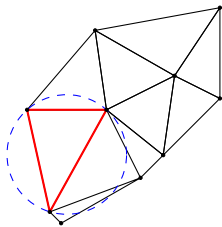


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Delaunay condition

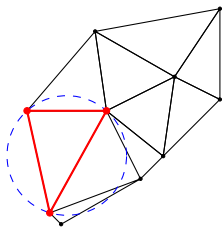
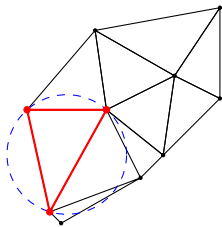


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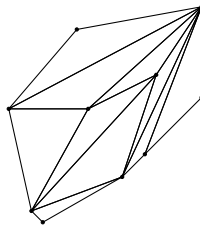
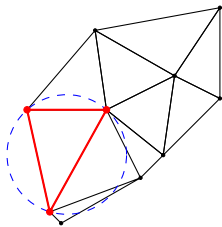
Delaunay condition



Respected

Figure: Example for Delaunay condition

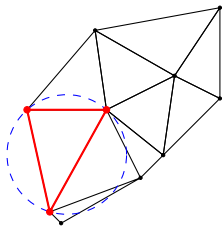
Delaunay condition



Respected

Figure: Example for Delaunay condition

Delaunay condition



Respected

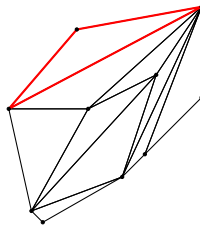
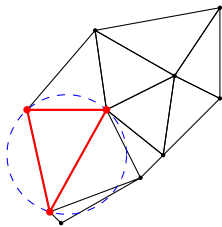


Figure: Example for Delaunay condition

Delaunay condition



Respected

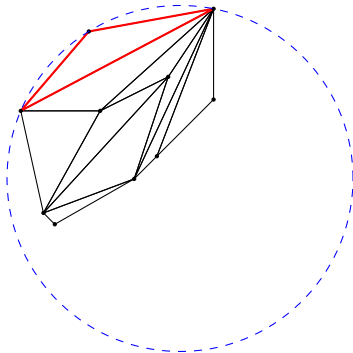
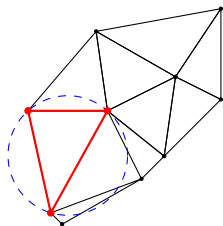


Figure: Example for Delaunay condition

Delaunay condition



Respected

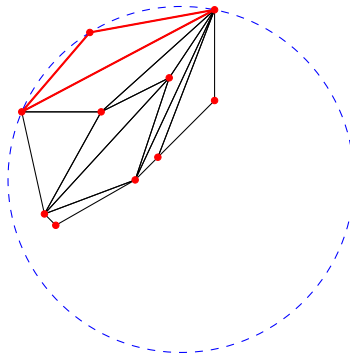
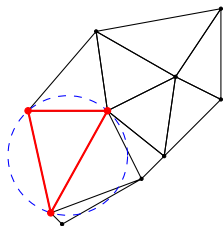
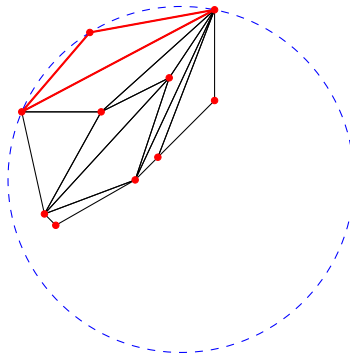


Figure: Example for Delaunay condition

Delaunay condition



Respected



Not respected

Figure: Example for Delaunay condition

Let's add some constraints

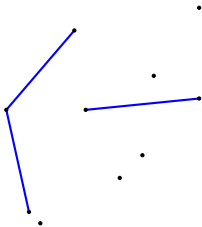


Figure: Edge constraints for Delaunay triangulation

Let's add some constraints



Figure: Edge constraints for Delaunay triangulation

Let's add some constraints

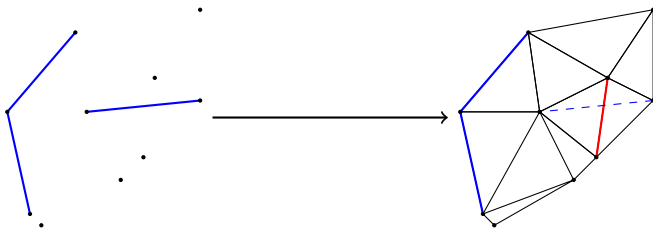


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Let's add some constraints

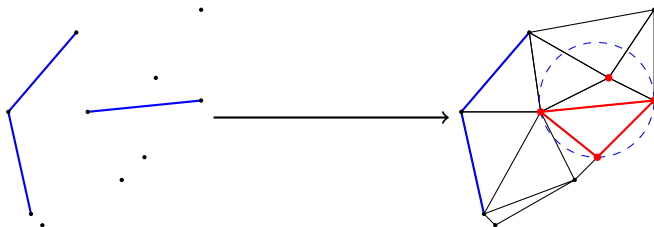


Figure: Edge constraints for Delaunay triangulation

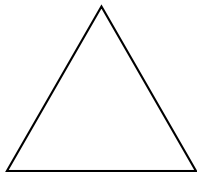
Let's add some constraints



Figure: Edge constraints for Delaunay triangulation

Tries to be "as good as possible"

Differences from 2D to 3D?



From this ...

Differences from 2D to 3D?



From this ...

... to this

Differences from 2D to 3D?



From this ...

... to this

Only ??

Not really...

... it gets harder

Not really...

... it gets harder

- Slivers appear

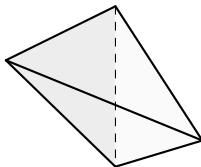


Figure: Example of sliver

Not really...

... it gets harder

- Slivers appear

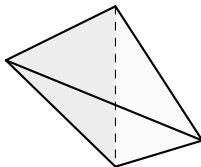


Figure: Example of sliver

- Constrained Delaunay tetrahedrisation are not always possible

TetGen [Si, 2015]

Already existing software

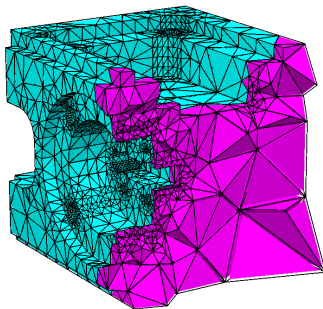


Figure: A model tetrahedrized by TetGen

- Not so robust, crashes on some models

Tetrahedral meshing in the wild (Tetwild)[Hu, 2018]

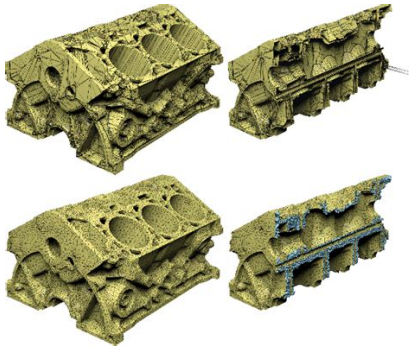


Figure: A model tetrahedrized by Tetwild

- Not strict on boundary

References



H. Si.

TetGen, a Delaunay-Based Quality Tetrahedral Mesh Generator.

ACM Trans. on Mathematical Software, 41 (2), Article 11, 2015.



Y. Hu, Q. Zhou, X. Gao, A. Jacobson, D. Zorin, D. Panozzo.
Tetrahedral Meshing in the Wild.

ACM Trans. Graph. 37, 4, Article 60, 2018.